# Editorial- Prime Maze

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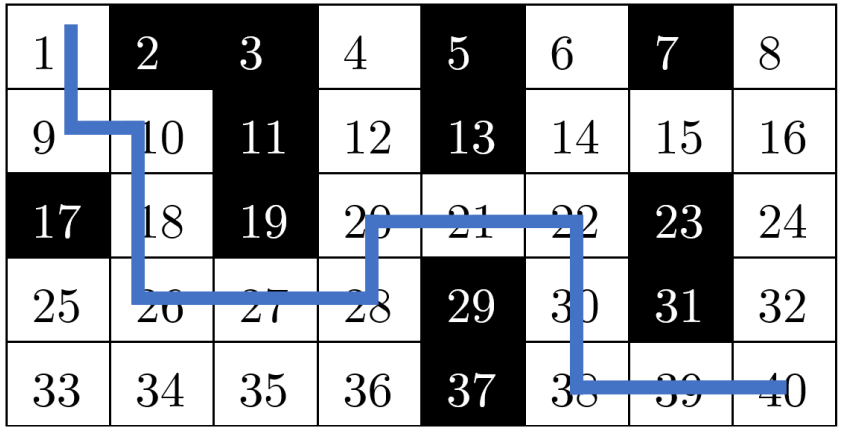
# Problem Statement

PDF Link:
https://drive.google.com/file/d/1punGJKb_BGUvtI7WBwMiRUJCkD4ZY6cn/view?usp=sharing

- Navigate a $R \times C$ maze with prime numbers being the obstacles.
- You can use a **move** to travel to adjacent blocks.
- Calculate the smallest number of move(s) required to travel from $(1, 1)$ to $(R, C)$.

- **Multiple test cases** $(T \leq 10^6)$.
- **Constraints**: $R \leq 2^{31}-1, C \leq 20$.

**$R$ is huge, $C$ is very small...**



# Subtasks (Invisible)

- **Subtask 1** (10 pts): $R \leq 5, C \leq 5$
- **Subtask 2** (20 pts): $T \leq 400, R \leq 20$
- **Subtask 3** (10 pts): $T \leq 1000, R \leq 1000$
- **Subtask 4** (60 pts): No further constraints

# Statistics

- **Max**. $30$ pts by **R8G1(PCMS)**
- **Min**. $10$ pts

- Expected score: $40$ pts by **exhaustion**

# Subtask 1

- Idea: How to get some score if you don't even understand the problem??
- Output `-1` !

- ☹☹ However, the test data includes a special case where $R = 1, C = 1$, and the program should answer this kind of query with `0` instead.

# Subtask 2&3

- Hey, we can use **breadth-first search**!
- Time complexity: $O(TRC)$
- **Not depth-first search!** We are looking for the smallest moves.

- Distinction between **DFS** and **BFS**:

DFS is usually for obtaining every possible paths by trial-and-error.

BFS is aimed expanding the states one at a time, so that the states using smaller moves will always be visited once and always the first.

# Subtask 2&3

**Procedures:**

1. Use Prime Sieve to obtain all prime numbers within the $R \times C$ range
2. Construct the maze explicitly
3. Apply the BFS algorithm

Don't forget to memset/clear your arrays ☺

# Subtask 4

**Strategies required:**

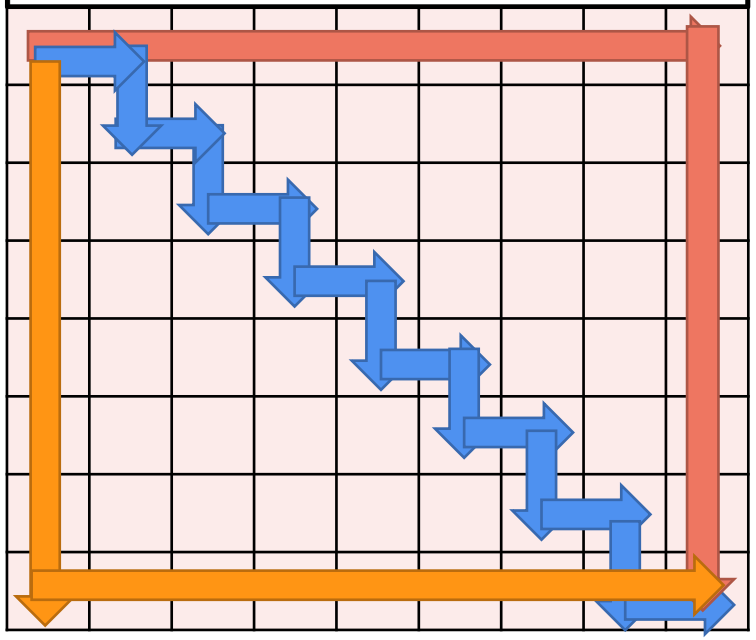
- Trial-and-error
- Observations and proving
- Classification
- Combining the subtasks together

# Subtask 4

- Solving Subtask 2&3, you now have a nice and clear solution for small test cases!
- Consider a $R \times C$ maze without any obstacles:

Example:

$R = 8, C = 9$



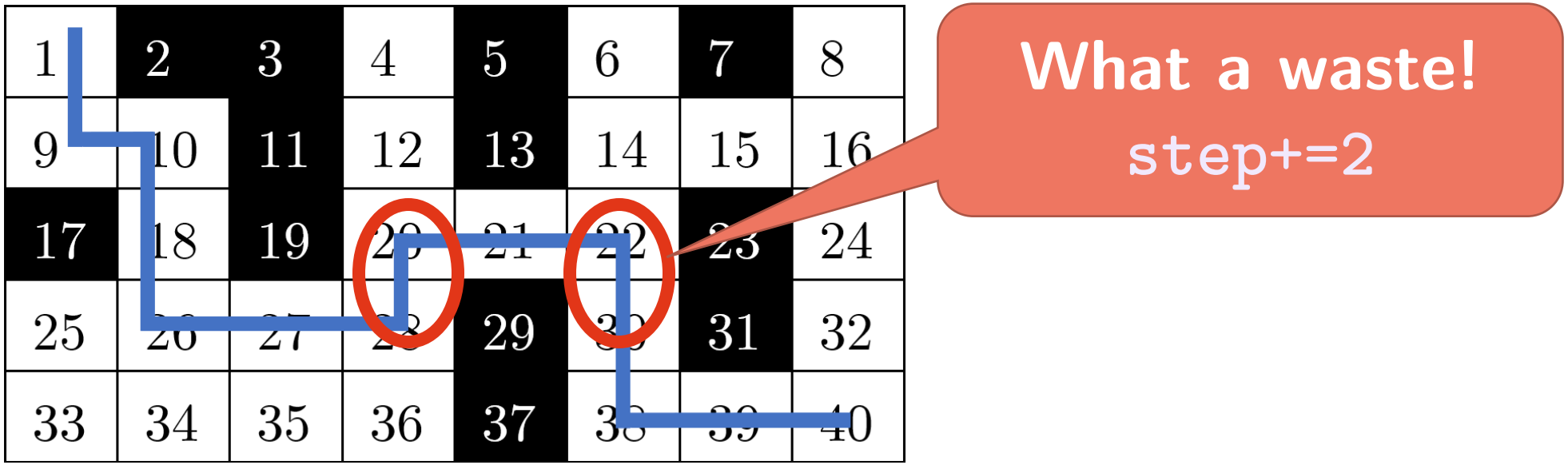
Smallest moves:

$$|8 - 1| + |9 - 1| = 15$$

Paths colored **red**, **blue** and **orange** are some possible paths with smallest moves.

# Subtask 4

- **What will cause the answer to increase?**
- Hints already provided in the statement!



# Subtask 4

**Key Observation 1**

The **last block in a row (except the first)** is always ***accessible***.

*Proof.* The number in block $(r, C)$ is $(r-1) \times C + C = r \times C$ (where $r > 1$), implying that this block is always accessible as it contains a composite number.

- Hence, we just need to find a way to connect the two columns together.
- Given $C$, the numbers that appear on the $C^{th}$ column are "fixed".
- Increasing $R$ can be seen as revealing the numbers in a row-by-row fashion.

# Case #1: C=1

- Let's just print the maze out and observe!

First 5 lines:

Example:
$R = 5, C = 1$

1
2
3
4
5

**Conclusion**:
Solvable if and only if $R = 1$
Else output "-1"

# Case #2: "I'm stuck!"

## Key Observation 2

No path exists as the maze is completely blocked by obstacles when $C = \mathbf{2,3,4,5,6,7,9,10,11,12,13,15,16,18.}$

Example 1:
$R = 5, C = 2$

| | |
|---|---|
| 1 | 2 |
| 3 | 4 |
| 5 | 6 |
| 7 | 8 |
| 9 | 10 |

Example 2:
$R = 5, C = 3$

| | | |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |
| 10 | 11 | 12 |
| 13 | 14 | 15 |

Example 3:
$R = 5, C = 7$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 8 | 9 | 10 | 11 | 12 | 13 | 14 |
| 15 | 16 | 17 | 18 | 19 | 20 | 21 |
| 22 | 23 | 24 | 25 | 26 | 27 | 28 |
| 29 | 30 | 31 | 32 | 33 | 34 | 35 |

**Conclusion**:
Output "-1"

# Case #3: Let's find the "winning strategy"

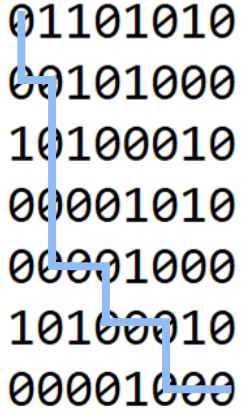
## Key Observation 3

When $C = \mathbf{8,14,17,19,20}$, we can always find a path connecting the first and the last column without "wasting" any moves, provided that $R \geq R_0(C)$, where $R_0(C)$ is a small constant that varies depending on $C$.

Example 1:
$R_0 = 7, C = 8$

```
Case: r=7, c=8
01101010
00101000
10100010
00001010
00001000
10100010
00001000
```

Strategy:



Other value of $R_0$s:
For $C = 14, R_0 = 8$
For $C = 17, R_0 = 11$
For $C = 19, R_0 = 10$
For $C = 20, R_0 = 11$

**Conclusion**:
When $R \geq R_0$, print the taxicab distance between $(1,1)$ and $(R,C)$

# Case #4: Cases left when R is small

**Key Observation 4**

When $C = \mathbf{8,14,17,19,20}$, and $R < R_0(C)$, the answer may be larger than $R + C - 1$.

- Generic and small enough, try using **BFS**!
- Use a **table** to avoid repetitive computations!

**Conclusion**: ☺
Combine the solutions together!

# Possible Errors and Solutions

| Error | Expected score | Solution |
| --- | --- | --- |
| 1. Missing the case where $R = 1, C = 1$ | 80 (**WA** on test case 1,3) | Output `0` when $R = 1, C = 1$ |
| 2. Using BFS every time solving subtask 4 when $R < 20$ | 90 (**TLE**) | Save the table beforehand and do $O(1)$ *queries* |
| 3. Not using `long long` (**$R + C - 2$ could be exceeding `int`**) | 90 | Change variable declarations to `long long` |